

Binary Tree Traversals

DAY 166
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In order to process trees, we need a Mechanism for traversing them, and that forms the Subject of this Section

The process of visiting all nodes of a tree is called tree traversal.

Each node is processed only once But it may be visited more than once. As we have already seen in linear data structures like linked lists, stacks, queues etc

the elements are visited in Sequential order. But in tree structures there are Many different ways.

Tree traversal is like Searching the tree, except that in traversal the goal is to move through the tree in a particular order. In addition, all nodes are processed in the traversal But Searching stops when the required node is found.

Traversal Possibilities

Starting at the root of a Binary tree, there are three main Steps that can be performed and the order in which they are performed defines the traversal type.

These Steps are : performing an action on the current node (referred to as "visiting" the node and denoted with "D") traversing to the left child node (denoted with "L") and traversing to the right child node (denoted with "R")

This process can be easily described through recursion, Based on the above definition there are 6 possibilities

- 1 **LDR** Process left Subtree, process current node data and process right Subtree.
- 2 **LRD** Process left Subtree, process Right subtree and then process the current node data.
- 3 **DLR** Process the current node data, process left subtree and then process right Subtree.
- 4 **DRL** Process the current node data, process right subtree and then process left Subtree
- 5 **RDL** Process right Subtree, process the current node data and then process left Subtree
- 6 **RLD** Process Right Subtree, process left Subtree and then process the current node data.

Classifying the Traversals

The Sequence in which these Entities (nodes) are processed defines a Particular Traversal method. The classification is Based on the order in which current node is processed.

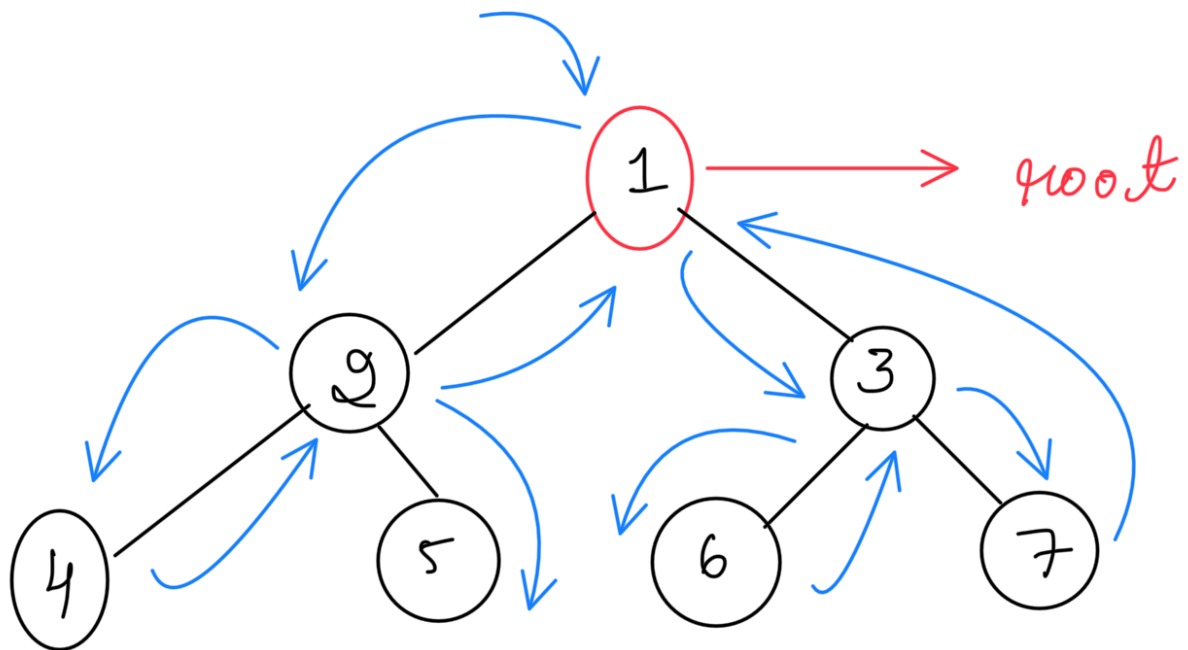
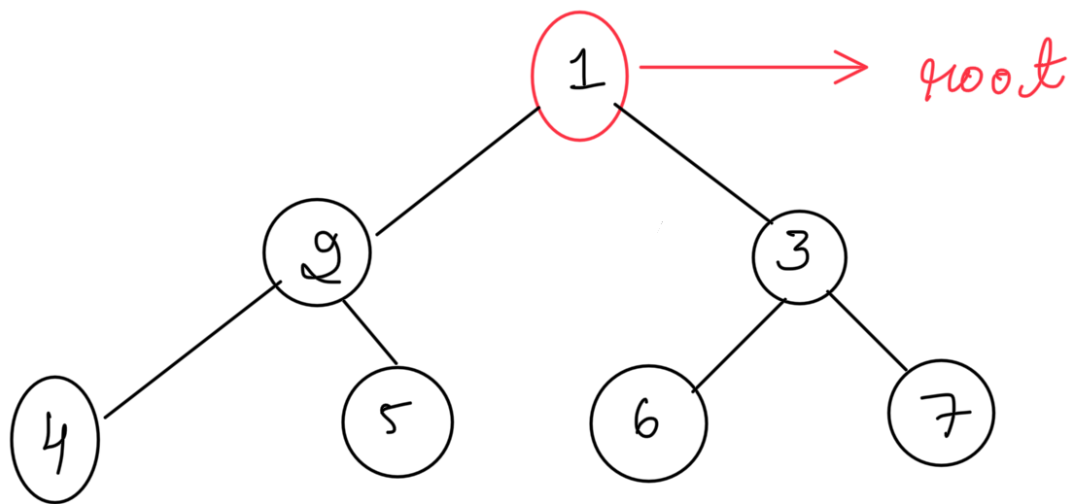
That means, if we are classifying Based on current node (D) and if D comes in the middle then it does not matter whether L is on left Side of D or R is on left of D

Similarly, it does not matter whether L is on right Side of D or R is on right Side of D. Due to this, the total 6 Possibilities are Reduced to 3 and these are.

- PreOrder (DLR) Traversal
- InOrder (LDR) Traversal
- PostOrder (LRD) Traversal

There is another Traversal method which does not depend on the above orders and it is.

- **Level Order Traversal**: This method is inspired from Breadth first Traversal (BFS of Graph algorithm)



Traversing →

Pre Order 1 , 2 , 4 , 5 , 3 , 6 , 7

In Order 4 , 2 , 5 , 1 , 6 , 3 , 7

Post Order 4 , 5 , 2 , 6 , 7 , 3 , 1